MIL-M-45577B(AL)
5 February 1985
SUPERSEDING
MIL-M-45577A(WC)
9 April 1976

MILITARY SPECIFICATION

MILLING MACHINE, SWIVEL HEAD, RAM TYPE, UNIVERSAL

This specification is approved for use by the US Army Armament Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of The Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a ram type, swivel head, universal milling machine. The machine is an electric motor driven, floor mounted model with equipment and accessories.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CC-M-1807	-	Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).

MILITARY

MIL-T-152	-	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173	-	Varnish, Moisture-and-Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-M-18058	-	Machinery, Metal and Woodworking, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.
AGMA 390.03 - Gear Handbook, Volume 1, Gear Classification, Materials, and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, 1901 N. Fort Myer Dr., Arlington, VA 22209.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slot; Their Bolts, Nuts, and Tongues.
ANSI B5.18 - Spindle Noses and Tool Shanks for Milling Machines.
ANSI B11.8 - Safety Requirements for the Construction, Care and Use of Drilling, Milling and Boring Machines.
ANSI IEEE 268 - Metric Practice.

(Application for copies should be addressed to the American National Standard Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.
ASTM E380 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches.
ISO 1122/1 - Glossary of Gear Terms - Part 1: Geometrical Definitions.

(Application for copies of ISO documents should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Metalworking Machine Tools and Plastics Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete milling machine for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.2 Design. The milling machine shall be new and of the manufacturer's current design meeting the requirements specified herein. The machine shall be designed to allow maintenance without using special tools. Parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, and replacement without modification, distortion, or damage to the machine or any machine part or component.

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* 3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the machine. In this specification, all measurements, dimensions, sizes, and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ANSI IEEE 268 or ASTM E380, as applicable.

* 3.2.2 Measuring and indicating devices. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI systems of measurement (see 3.3.5.1). When a dual US and SI system is furnished, feed dials shall have independent zero adjustments and shall be calibrated so that the last dial graduation progresses into, and is continuous with, the first dial graduation as the dial is rotated through the zero position. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). regardless of the measurement system used, all measuring and indicating devices on the milling machine shall be graduated in the same system.

* 3.2.3 Safety and health requirements. Covers, guards, and other safety devices shall be provided. The devices shall not interfere with operation of the machine. The safety devices shall prevent unintentional contact with the guarded part and shall be removeable to facilitate inspection, maintenance, and repair of machine components. All machine parts, components, mechanisms, and assemblies of the milling machine, whether specifically required herein or not, shall comply with those requirements of OSHA 2206 and ANSI B11.8 applicable to milling machines. In the event of conflict between the requirements, the OSHA standards shall apply. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

* 3.2.4 Lubrication. All bearings (except sealed, permanently lubricated type), mating gears, and all other moving parts shall be provided with means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. All oil holes, grease fittings, and filler caps shall be accessible. The manufacturer shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practices of the machine tool industry. The lubrication systems shall be temporarily capped or stopped to prevent loss of lubricants during shipment. The manufacturer shall include a warning notice in the instruction manual to remove the temporary caps and stoppers to prevent the start-up or use of the milling machine without removal.

* 3.2.5 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

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* 3.3 Construction. The milling machine shall be constructed of parts and materials which are without defects and are free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum rating and capacity without distortion. All parts and mechanisms subject to periodic adjustment, repair, or replacement shall be accessible without distorting, modifying, or damaging the machine or any part or component. All screws, pins, and similar parts shall be installed to prevent loss of tightness. Construction of the machine shall be such that, when installed and connected to the proper power supply, the machine shall be ready for operation.

* 3.3.1 Castings and forgings. Castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

* 3.3.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations shall not be employed as repair measures for defective parts.

* 3.3.3 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

* 3.3.4 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.3.5 Dials and plates.

* 3.3.5.1 Dials. All dials shall be graduated in either or both the US customary inch and the Metric (SI) system of measurement (see 3.2.2). Graduations indicating stock removal or tool movement shall be in increments of not more than 0.001 inch and 0.02mm. Dial diameters shall be such that the graduations may be read from the machine operator's normal working-height position. Dials shall be permanently and legibly etched or engraved on a non-glare background. The dials shall be accurate indicators for the adjustment of machine or machine component movement. The dials shall be provided with a means for locking at any desired setting.

* 3.3.5.2 Instruction and indicating plates. All words on instruction and indicating plates shall be in the English language. Characters shall be permanently and legibly engraved, etched, embossed, or stamped in bold face on a contrasting background.

* 3.3.6 Gears. All gears and pinion designs selected for use in the milling machine or its components shall be fabricated to either English (USA) standard inch diametral pitch per AGMA 360.02 and AGMA 390.03 or Metric (SI) module ratio per the International Organization for Standardization (ISO) ISO 54 and ISO 1122/1 and ASTM E380.

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* 3.3.7 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the machine tool industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.3.7.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.4 Components. The milling machine shall be composed of, but not limited to, the components specified in 3.4.1 thru 3.4.7.

3.4.1 Base and column. The base and column shall be of such design and weight to assure the accuracies and performance specified herein with a minimum of vibration. The base shall have provision for leveling and securing the machine to the floor or foundation. The column shall have machined and fitted ways for vertical travel of the knee. Provision shall be made for mounting the ram to the top of the column.

* 3.4.2 Knee. The knee shall be of rigid construction with internal ribbing and shall support the saddle, table, and related feed mechanisms. Means shall be provided to clamp the knee in any desired position on the column. Wipers or other suitable means shall be provided to prevent dirt and chips from entering ways. The knee elevating screw shall have either square or acme threads and shall be completely enclosed in a telescopic or accordian type guard.

3.4.3 Universal saddle. The universal saddle shall be of rigid construction having machined ways provided with guides and adjustment gibs. Wipers, or other suitable means, shall be provided to prevent dirt and chips from entering ways. An indexing scale shall be provided on the saddle to indicate swivel setting of the table. Clamps shall be provided to secure the saddle in any swiveled position. The length of the ways shall be not less than the table travel.

3.4.4 Table. The table shall be of rigid construction, capable of supporting work under cutting loads. The top surface of the table shall be machine finished. Not less than three equally spaced T-slots conforming to ANSI B5.1, the size specified in Table I, shall be provided. The front side of the table shall be machine finished and provided with means for mounting adjustable table stops. A full length adjustable gib shall be provided for the table. Wipers, or other suitable means, shall be provided to prevent dirt and chips from entering ways. The table feed screw shall have either square or acme threads. The table shall have coolant drain channels with chip screens and provision for return of coolant to reservoir.

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TABLE I. Sizes and capacities.

Characteristics		Minimum requirements
Table working surface		50 (long) by 10 (wide)
Longitudinal travel		27
Saddle (cross travel)		10
Knee (vertical travel)		16
Table (swivel adjustment)		45° swivel to left of zero, 35° to right of zero
Ram travel		19
Spindle to table (horizontal position)		17 inch distance from spindle center line to table top (knee in lowest position)
Spindle head swivel		90° (vertical to horizontal)
Spindle nose taper		No. 40, ANSI B5.18
Spindle speeds		14
Speed range		80 to 1500 revolutions per minute (rpm)
Number of feeds		12
Feed rate - Longitudinal		0.5 to 20 7/8 inches per minute (ipm)
	Cross	0.5 to 20 7/8 ipm
	Vertical	0.25 to 10 7/16 ipm
Power rapid travel -	Longitudinal	85 ipm
	Cross	85 ipm
	Vertical	42 1/2 ipm
Motors - Spindle		3 horsepower (hp)
Feed		1 hp
Pump		1/8 hp
T-Slots - Number		3
Width of throat		11/16

NOTE: Data presented in inches, unless otherwise stated.

3.4.4.1 Limit stops. Adjustable trip dogs for automatic kickout of power feeds shall be provided. Positive limit stops shall be provided to prevent damage to the machine in case of overrunning feeds or rapid travel in all six directions.

3.4.5 Ram. The ram shall be a metal casting mounted on top of the milling machine column. A rack and pinion shall be provided for moving the ram in a horizontal direction. Provision shall be made for clamping the ram in any position within its range of travel.

3.4.6 Swivel head. An adjustable swivel head shall be attached to the ram. The swivel head shall be capable of being rotated through 90° from the vertical to horizontal position, perpendicular to the work table. A scale, graduated in degrees, shall be provided to indicate the true position of the swivel head. The swivel head shall be equipped with positive locating stops and clamping device. The spindle nose specified in Table I shall be provided as an integral part of the swivel head.

* 3.4.7 Coolant system. The coolant recycling system shall include a sump or reservoir, an electrically power driven pump, and all necessary piping. The sump or reservoir shall be of sufficient capacity to permit the continuous flow

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of coolant while maintaining at least 7 gallons of coolant in the sump or reservoir. The system shall have a means for draining and cleaning. The system shall include a strainer which can be removed for cleaning. A valve and flexible metal hose, conveniently located near the machine operator's normal working position, shall be provided to permit the operator to direct and control the amount of coolant flow.

3.5 Sizes and capacities. The milling machine shall conform to the dimensions, capacities, and requirements set forth in Table I. Where a range is shown, the milling machine shall include the entire range.

3.5.1 Speeds. The speed range and speed settings shall meet or exceed the requirements listed in Table I. Means shall be provided for selecting the various speeds. A dial or plate shall be provided to indicate the speed in use. The spindle speeds shall be independent of the table feeds.

3.5.2 Feeds. Each longitudinal, transverse, and vertical movement of the table, saddle, or knee shall have hand and power feed in two directions. The feed range and power feed rates shall be as specified in Table I. Feed rate selection shall be made at the front or side of the milling machine. Dials or plates shall be provided to indicate the feed rate in use. Levers used for engaging the table, saddle, and knee shall operate independently of one another in two directions and in neutral. The direction of the table lever shall indicate the direction of table feed. Adjustment of the table, saddle, and knee shall be indicated by micrometer dials, graduated in increments of 0.001 inch or less. The table, saddle, and knee shall be provided with locking devices for any desired position. All hand adjusting wheels or cranks shall be mounted in a manner to prevent them from revolving when power feeds are engaged.

3.5.3 Rapid travel. Power rapid travel shall be provided in six directions, selected by the power feed engaging levers, and controlled by a single hand operated lever located in the front part of the machine. The engaging movement of the rapid travel lever shall be in a direction that reduces the possibility of accidentally engaging the rapid travel. Rapid travel shall be at a constant rate not less than the speeds specified in Table I, and shall be independent of spindle speeds and table feed rates. Rapid travel shall be available at all times the milling machine is running, whether the spindle is revolving or stationary.

3.6 Performance, accuracy, and alignment. The milling machine shall be capable of meeting or exceeding the requirements of 3.6.1 thru 3.6.3.

3.6.1 Performance. The milling machine shall be used to perform milling operations on a piece of 1045 steel, which is 3 inches wide by 2 inches thick by 12 inches long. The stock removal rate shall be not less than 0.65 cubic inches per minute per rated horsepower of the spindle drive motor.

3.6.2 Accuracy. The milled surfaces shall have a finish of 125 microinches arithmetical average (aa) or better. The milled surfaces shall be flat within 0.001 inch, total deviation of the entire surface. Adjacent sides shall be square within 0.001 inch per foot. Opposite sides shall be parallel within 0.001 inch per foot.

3.6.3 Alignment tolerance. The milling machine shall not exceed the alignment tolerances of Table II.

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TABLE II. Alignment tolerances.

Characteristics	Maximum deviation (inches)
Spindle end play	.0005 tir (total indicator reading)
Spindle run out	.0005 tir
Transverse rise and fall of table	.0007 per foot
Longitudinal rise and fall of table	.007 per foot
Transverse table travel square with longitudinal table travel	.001 per foot
Knee travel square with table	.0005 per foot
Spindle square with table surface	.001 tir
T-slots parallel to table travel	.001 per foot

* 3.7 Electrical system. The electrical system shall conform to applicable requirements of NFPA 79. A disconnect switch shall be provided to disengage all incoming power and shall be separate from the on-off switch of the machine. The machine shall be capable of operating on 230/460 volt, three phase, 60 hertz power (initially wired to operate on 230 volts). An identified terminal shall be provided, suitable for connecting the proper size grounding conductor in accordance with the requirements for the specified power source.

3.7.1 Motors. Motors shall be continuous duty, with motor hp conforming to the requirements shown in Table I. Motors shall be induction squirrel-cage type, 230/460 volts, three phase, 60 hertz power (initially wired for 230 volts). Unless otherwise specified (see 6.2.1), motors shall be single speed, shall have a drip-proof enclosure, and shall comply with CC-M-1807.

3.8 Standard equipment. The following standard equipment shall be furnished with each machine:

One set of wrenches, pulleys, belts, and guard for spindle drive, as applicable.
One arbor support, style "A" with 23/32 inch hole.
One arbor support, style "B" for 2 1/8 inch sleeve.
One outer brace, as applicable.
One draw-in bar, as applicable.
One adapter, collet.
One collet, 7/8 inch diameter.

3.8.1 Spindle adapter(s). If spindle tapers furnished on the milling machine are not identical, adapter(s) shall be furnished to interchange between the different tapers.

3.9 Additional equipment. The following additional equipment shall be furnished with each machine.

3.9.1 Arbors.

One arbor, 7/8 inch diameter by 10 inches long.
One arbor, 1 inch diameter by 15 inches long.
One arbor, fly cutter.
One arbor, shell end mill for 3/4 inch arbor hole.
One arbor, shell end mill for 1 inch arbor hole.
One arbor, shell end mill for 1 1/4 inch arbor hole.

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3.9.2 Collets.

Three collets, 11/16 inch through 13/16 inch, by 16ths.
Eighteen collets, 1/8 inch through 21/32 inch, by 32nds.

3.9.3 Chuck. A six inch diameter, three-jaw universal chuck, capable of being mounted on a dividing head, shall be provided. The scroll plate and jaws shall be made of heat treated alloy steel. The permissible error of the chuck shall not exceed the tolerances contained in Table III.

TABLE III. Permissible error.

Characteristics	Maximum deviation (inches)
Taper of dividing head spindle runs true:	
a. At a distance 1/2 inch from nose	.0002 inch tir
b. At a distance 12 inches from nose	.0008 inch tir
Point of center runs true	.0002 inch tir
Dividing head spindle square with table top	.0008 inch per foot
Dividing head spindle parallel with center T-slot	.0004 inch per foot
Dividing head spindle directly over center of center T-slot (lateral, off-set)	.0004 inch per foot
Maximum permissible dividing error:	
a. For intermediate divisions	±20 seconds
b. For the total	±1 minute
Working axis (mandrel between centers) parallel with table top (degree of incline between dividing head and tailstock)	.0008 inch per foot
Working axis (mandrel between centers) parallel with center T-slot (lateral off-set between dividing head and tailstock)	.0008 inch per foot
Run-out of chuck at six inches from spindle nose using standard ground bars	.003 inch tir

3.9.4 Dividing head. The dividing head provided shall swivel in the vertical plane, have a minimum swing of ten inches, and be complete with center, index plates, adjustable tailstock, and center rest. At least four T-bolts conforming to ANSI B5.1, with nuts and washers, shall be furnished with each dividing head. The dividing head spindle shall be mounted on bearings. The spindle nose shall be at least No. 40, conforming to ANSI B5.18. Provision shall be made for taking up wear between the worm and worm gear. Adjustment shall be through an arc greater than 90°. Provision shall be made for locking the head in any of its positions. The accuracy of the dividing head between indexes shall be within 0.0005 inch. The accumulative error of the index head shall not exceed 0.0015 inch when measured on a circle with a maximum diameter of ten inches. Unless otherwise specified (see 6.2.1), index plates shall provide for indexing all divisions from one to at least 50 and all even divisions to at least 100 with all numbers divisible by five up to 520. The permissible error of the dividing head shall not exceed the tolerances specified in Table III.

* 3.9.5 Swivel vise. The swivel vise shall have a full 360 degree range and shall be graduated in increments of one degree. The vise opening shall have a range of zero inches to five inches or more. Jaws shall be hardened, ground, and removable. The vise shall be complete with at least two T-bolts conforming to ANSI B5.1, with nuts, washers, and crank.

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* 3.10 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

* 3.11 Lubrication chart or plate. A corrosion-resistant metallic lubrication plate, with permanently legible English language characters, shall be attached to each machine in a location visible to the machine operator. If a chart is furnished, it shall be placed in a transparent plastic folder or laminated between clear plastic permanently sealed sheets with suitable means for mounting. The following information shall be furnished on the chart and plate:

Points of application.
Service interval.
Type of lubricant.
Viscosity.
Military or Federal specification number of lubricant when applicable.

* 3.12 Nameplate. A corrosion-resistant metallic nameplate, with permanently legible English language characters identifying the machine in accordance with MIL-STD-130, shall be attached to each machine in a location visually convenient to the machine operator. The following information shall be included on the nameplate:

National Stock Number.
Manufacturer's name.
Manufacturer's model number.
Manufacturer's serial number.
Contract number.
Date of manufacture.

* 3.13 Workmanship. Standards of workmanship shall assure that the milling machine shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Tests (see 4.6).
(c) Packaging inspection (see 4.7).

* 4.4.1 Sampling. When required (see 6.2.1), sampling for quality conformance inspection shall be as specified in MIL-STD-105 at the Inspection Level and Acceptable Quality Level (AQL) specified in the contract. For packaging inspection (see 4.7), special inspection level S-4 and an AQL of 4.0 percent defective shall be used.

* 4.5 Product examination. The milling machines shall be examined to determine compliance with the requirements of Section 3 and the contract. As a minimum, the following characteristics shall be examined:

CHARACTERISTICS	REQUIREMENT PARAGRAPH
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Nameplate	3.12
Workmanship	3.13

A certificate of conformance may be used to satisfy examination requirements when more than a visual or dimensional verification would be required.

4.6 Tests.

4.6.1 Operational test. The milling machine shall be operated at no load for not less than 30 minutes. Operation of all controls, motors, adjusting mechanisms, and other accessories shall be inspected during this test period.

4.6.2 Alignment tolerance test. The milling machine shall be tested for compliance with the alignment requirements of Table II (see 3.6.3).

4.6.3 Performance test. The milling machine shall mill a work piece of 1045 steel as specified in 3.6 thru 3.6.2. The work piece shall not be less than 12 inches long by 3 inches wide by 2 inches thick. One of the two surfaces having the largest area shall be ground flat and mounted in contact with the table. The work piece may be drilled to facilitate clamping. The milling machine shall make a roughing cut to mill metal off the entire horizontal surface of the work piece in one pass. Sufficient depth and feed rate to load the spindle drive motor to not more than its rated hp shall be used. The milling machine shall be operated at a spindle speed not exceeding 10 percent of the highest speed required in Table I. The resultant metal removal rate shall not be less than 0.65 cubic ipm per rated hp of the spindle drive motor. The surface finish of the work piece after machining shall not be rougher than 125 microinches aa. A roughing cut shall be made on a vertical face of the work piece. Finishing cuts shall be made on the horizontal and vertical faces which were previously rough cut. The test allowances of the work piece after the finishing cuts shall be as specified in 4.6.3.1, 4.6.3.2, and 4.6.3.3.

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4.6.3.1 Squareness. The adjacent milled sides shall be square within 0.001 inch per foot.

4.6.3.2 Flatness. The two milled surfaces shall be inspected for flatness. Each surface shall be flat within 0.001 inch, total deviation of the entire surface.

4.6.3.3 Parallelism. The two finished paralleled sides shall be parallel within 0.001 inch per foot.

* 4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

* 5.1 Military preservation, packing, and marking. When specified (see 6.2.1), requirements for Level A preservation and for Levels A and B packing and marking shall be in accordance with MIL-M-18058.

* 5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

6. NOTES

6.1 Intended use. The milling machine covered by this specification is intended for use in tool rooms, machine shops, and laboratories.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Measuring and indicating device graduation(s) required (see 3.2.2 and 3.3.5.1).
(d) Exceptions and additional safety/health requirements when required (see 3.2.3 and 6.4).
(e) Painting if different (see 3.3.3).
(f) Motor if different (see 3.7.1).
(g) Index plates if different (see 3.9.4).
(h) First article inspection if different (see 4.3).
(i) Quality conformance inspection if different (see 4.4).
(j) Sampling when required (see 4.4.1).
(k) Level of preservation, packing, and marking if different (see 5.1 and 5.2).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

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6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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